

MAJOR PROJECT REPORT

Obstacle Avoiding Robot

Autonomous Robot Using Arduino Uno, L293D Motor Driver and Servo-Mounted Ultrasonic Sensor

PROJECT CATEGORY Major Project 1 - Obstacle Detection Robot	PROGRAM IoT & Robotics Training Program
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Student details Name: _____ Enrollment No.: _____

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Declaration

This report documents the design, wiring, program logic and planned verification of an autonomous obstacle-avoiding robot. The student should complete the name fields and attach final hardware photographs, demonstration video link and GitHub repository link before submission.

Prepared in accordance with the project-report requirements: title, objective, description, component list, circuit diagram, circuit explanation, hardware design, software implementation, workflow, testing, conclusion and learning outcomes.

1. Project Objective

To design and implement a mobile robot that continuously measures the distance to objects in front of it and autonomously avoids collisions. The robot stops when an obstacle enters its safety zone, reverses slightly, scans both sides using a servo-mounted ultrasonic sensor, and turns toward the side with more free space.

2. Project Description

The robot uses an Arduino Uno as its controller. A single-wire PING))) ultrasonic sensor provides distance information. The sensor is mounted on an SG90 servo so that the robot can inspect the left and right directions. An L293D H-bridge receives four Arduino control signals and drives two DC geared motors for forward, reverse, left, right and stop actions. Diagnostic measurements and decisions are printed to Serial Monitor at 9600 baud.

3. Mandatory Requirements Coverage

Requirement	Implementation in this project
Obstacle sensor	PING))) ultrasonic distance sensor; front, left and right readings.
Autonomous response	Stops, reverses, scans, selects a clearer path, and turns.
Motor system	Two DC geared motors driven independently through L293D.
Navigation feedback	Serial Monitor shows front distance, side distances, and chosen turn.
Clean code	Modular functions for measurement, scanning, motor actions and avoidance.
Testing	A physical verification checklist and calibration procedure are included.

4. Functional Behaviour

- When the path is clear beyond 25 cm, the robot drives forward.
- When an obstacle is detected at or inside 25 cm, the robot stops immediately.
- The robot reverses briefly to create clearance, then measures both sides.
- It turns toward the larger measured distance and resumes monitoring.
- If an ultrasonic reading times out, the robot stops instead of moving blindly.

Important operating note

The supplied diagram uses a rectangular 9 V battery. For a physical build, use a suitably rated rechargeable motor pack (typically 6-9 V) because common rectangular 9 V batteries often cannot deliver stable motor and servo current.

5. Component List

Component	Qty.	Purpose
Arduino Uno	1	Reads sensor data and controls the servo and motor-driver inputs.
L293D H-bridge motor driver	1	Provides directional motor control and isolates the Arduino from motor current.
DC geared motors with wheels	2	Provide differential-drive motion.
PING))) ultrasonic sensor	1	Measures obstacle distance using its single SIG pin.
SG90 micro servo	1	Rotates the sensor to scan left, centre and right.
Robot chassis and caster wheel	1 set	Provides mechanical structure and stable support.
Motor battery / supply	1	Powers the L293D motor supply and DC motors.
Breadboard and jumper wires	1 set	Creates the driver and control connections.

6. Pin Connections

Arduino pin	Connected module	Function
D8	L293D IN1	Left motor forward
D7	L293D IN2	Left motor reverse
D6	L293D IN3	Right motor forward
D5	L293D IN4	Right motor reverse
D3	PING))) SIG	Single-wire trigger/echo distance signal
D2	Servo signal	Sensor scan angle command
5V	Sensor, servo, L293D logic	Logic supply
GND	All modules and motor supply	Shared electrical reference

Connection checks before power-up

- Connect the Arduino ground, L293D ground, sensor ground, servo ground and motor-supply negative together.
- Keep motor power on the L293D motor-supply pin; do not power motors directly from an Arduino I/O pin.
- Verify L293D enable pins are HIGH as shown, and ensure the IC orientation matches the diagram.
- If a wheel rotates the wrong way, reverse that motor's two wires or swap its forward/reverse logic in the code.

7. Circuit Diagram

Figure 1 shows the circuit diagram supplied for this project. It includes the Arduino Uno, L293D motor driver, two DC geared motors, servo and single-wire ultrasonic sensor.

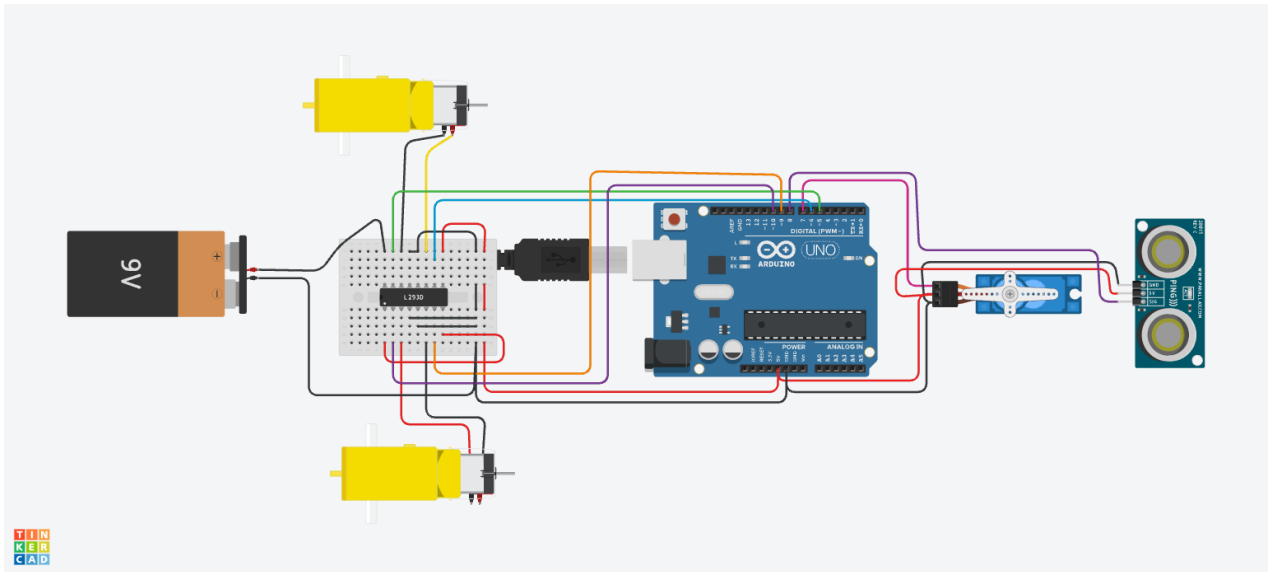


Figure 1. Obstacle-avoiding robot circuit diagram (source: supplied project image).

8. Circuit Explanation

The Arduino provides four digital control signals to the L293D. Each motor receives a forward/reverse pair; setting one input HIGH and the paired input LOW drives the motor in one direction, while reversing the pair reverses the motor. The code combines the two motor directions to make the robot move forward, reverse, pivot left, pivot right or stop.

The ultrasonic module shares one SIG pin with Arduino D3. The program sends a short trigger pulse, then measures the returned pulse width to calculate distance. The servo receives an angle command from D2 and positions the sensor at approximately 25 degrees (right), 90 degrees (centre) and 155 degrees (left).

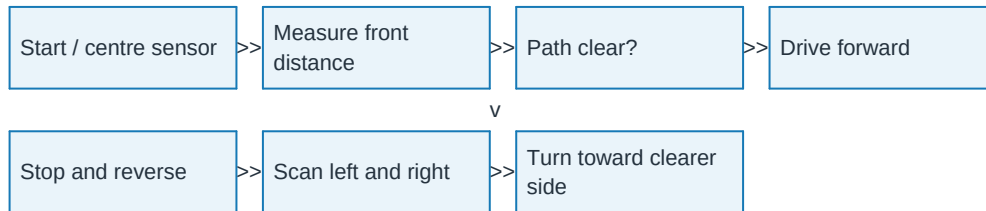
Power rule

The motor supply must share ground with the Arduino. Servo/motor current can cause resets if the supply is weak; use a stable, appropriately rated supply for the physical robot.

9. Hardware Design

The design uses a two-wheel differential-drive chassis with a caster wheel. The ultrasonic sensor is placed on the front servo horn so that it can observe the robot's path and compare both side directions. The Arduino and motor driver are mounted centrally to keep wiring short and the chassis balanced. Motors should be secured so that the left/right wheel directions match the program's forward command.

10. System Workflow / Navigation Algorithm



Algorithm steps

- Initialise all motor pins as outputs, attach the servo and centre the ultrasonic sensor.
- Measure front distance using a trigger pulse and pulse-width timing.
- If the measurement is invalid, stop the motors and wait for a valid reading.
- If front distance is greater than the safety threshold, drive forward.
- Otherwise stop, reverse briefly, measure right and left distances, then centre the sensor.
- Turn toward the side with the larger free-space reading, stop, and return to front monitoring.

Calibration parameters

Parameter	Default	Adjustment purpose
SAFE_DISTANCE_CM	25 cm	Increase for earlier avoidance; reduce only for slow, controlled robots.
Reverse delay	260 ms	Sets how much clearance the robot creates before scanning.
Turn delay	430 ms	Sets turn angle; tune according to wheel speed and floor grip.
Servo angles	25 / 90 / 155 degrees	Centres and aims the sensor toward clear left/right views.

11. Software Implementation

The source file **Obstacle_Avoiding_Robot.ino** is modular. Dedicated functions keep the program easy to test and modify: *readDistanceCm()* measures distance, *scanDistanceAt()* samples a side, motor functions control each movement, and *avoidObstacle()* performs the collision-avoidance sequence.

Pin definitions and navigation constants

```
const byte LEFT_MOTOR_FORWARD_PIN = 8;
const byte LEFT_MOTOR_REVERSE_PIN = 7;
const byte RIGHT_MOTOR_FORWARD_PIN = 6;
const byte RIGHT_MOTOR_REVERSE_PIN = 5;
const byte ULTRASONIC_SIGNAL_PIN = 3;
const byte SERVO_PIN = 2;
const int SAFE_DISTANCE_CM = 25;
```

Obstacle response

```
if (frontDistance > SAFE_DISTANCE_CM) {
  moveForward();
} else {
  avoidObstacle(); // stop, reverse, scan both sides, turn
}
```

Software setup

- Install Arduino IDE and open the supplied .ino file.
- Select Tools > Board > Arduino Uno, select the correct COM port, and upload.
- Open Serial Monitor at 9600 baud to view distance readings and turn decisions.
- If the physical wiring differs, change only the labelled pin constants at the top of the sketch.

Source-code note

This sketch targets the single-wire PING))) ultrasonic sensor shown in the circuit. A 4-pin HC-SR04 requires separate trigger and echo pin logic and should not be substituted without updating the measurement function.

12. Testing Results and Verification Checklist

Physical testing must be completed after assembly. The table records the expected observable behaviour and can be marked during the demonstration. It intentionally does not claim results that have not yet been measured on the final hardware.

Test case	Expected behaviour	Record after test
Clear path (>25 cm)	Robot drives forward; Serial Monitor prints front distance.	Pass / Fail: _____ Notes: _____
Front obstacle (<=25 cm)	Robot stops and reverses before choosing a turn.	Pass / Fail: _____ Notes: _____
More room on left	Servo scan shows larger left distance; robot turns left.	Pass / Fail: _____ Notes: _____
More room on right	Servo scan shows larger right distance; robot turns right.	Pass / Fail: _____ Notes: _____
Sensor timeout	Robot remains stopped and prints Sensor timeout message.	Pass / Fail: _____ Notes: _____
Different surfaces/objects	No collision after threshold and turn-delay calibration.	Pass / Fail: _____ Notes: _____

13. Challenges Faced and Mitigation

Challenge	Mitigation
Motors can rotate in opposite directions because of terminal orientation.	Swap a motor's two wires or its forward/reverse constants; retest forward movement.
Servo/motor load can cause Arduino resets on a weak battery.	Use a properly rated motor battery and keep all grounds common.
A fixed turn time changes with battery level and floor friction.	Tune the 430 ms turn delay on the actual chassis and test both directions.
Ultrasonic echoes can be unreliable on soft or angled surfaces.	Use timeout protection, test different objects, and select a conservative safety distance.

14. Conclusion

The project design meets the core obstacle-detection robot requirements by combining real-time ultrasonic sensing, servo scanning, differential motor control and an autonomous avoidance routine. With physical calibration and verification, the robot can demonstrate smooth collision avoidance across common indoor obstacles.

15. Learning Outcomes

- Interfacing an ultrasonic distance sensor and interpreting pulse width as distance.
- Controlling two DC motors through an H-bridge for differential-drive navigation.
- Using a servo to expand a sensor's field of view.
- Structuring Arduino code with readable, reusable functions and safety handling.
- Testing electromechanical systems and calibrating behaviour on real hardware.

Submission reminder: add completed hardware photographs, demonstration video link, GitHub repository link and commit history before emailing the final submission.